

Public CAD Is Not Operational Ground Truth

Regime Change, Hurricane Ida, and the Measurement of Police Responsiveness

25 August 2026

Abstract

Public computer-aided dispatch (CAD) records are often treated as operational clocks. New Orleans public CAD instead reveals a change in the released measurement product before Hurricane Ida. Among non-officer-initiated records, the state “arrival field present, dispatch field absent” is absent from 419,840 records through 27 July 2021, first appears on 28 July, and grows thereafter. A first-hand audit of the official July file confirms that the break is present in raw nonblank and parseable fields: officer-initiated dispatch-field presence simultaneously falls from 100 percent on 27 July to 40.0 percent on 28 July and 1.0 percent on 29 July. Five weeks later, Ida produces a further two-day reconfiguration. The maximum public-field contrast is rank 1 among 153 prespecified stage-era ordinary-week references and rank 1 among 151 support-independent post-change references. A 4,000-replicate full-count bootstrap places Ida first in every draw. The standardized analysis covers 86.1 percent of event records and 77.0 percent of baseline records and fails the reference windows’ 0.90 eligibility rule, so the full-count result is essential. Public institutional documents define field labels and exclude two proposed technology explanations, but no archived changelog or historical internal-to-public bridge identifies the July cause. In the comparable non-officer denominator, dispatch-field completeness is 65.4 percent in 2024, 66.0 percent in 2025, and 66.8 percent in the 2026 snapshot; the often-quoted 41.8 percent is the all-record rate, depressed by officer-initiated conventions. Public CAD supports reproducible administrative-field measures, not an automatic measure of physical dispatch, arrival, capacity, response quality, or domestic-violence incidence.

Keywords: calls for service; police responsiveness; administrative data; Hurricane Ida; measurement error; partial identification

JEL codes: C18, C55, H12, K42

1 Introduction

Public CAD files appear to record a direct sequence: a call is created, an officer is dispatched, and an arrival is entered. That sequence becomes a performance measure only if the released fields retain stable meanings, coverage, and lineage. A missing dispatch timestamp can reflect an operational event, a workflow exception, later entry, retention, redaction, or export logic. These pathways are observationally equivalent in a public file.

Write D_t^{CAD} for reported demand entering CAD, C_t for effective capacity, and P_t for the priority process. Operations produce a service path

$$S_t = g(D_t^{\text{CAD}}, C_t, P_t), \quad (1)$$

while recording, retention, and export R_t^{rec} produce the released record

$$Y_t = h(S_t, R_t^{\text{rec}}). \quad (2)$$

A public timestamp is a feature of Y_t , not proof of its latent counterpart in S_t .

Hurricane Ida makes this distinction visible, but the first empirical result predates the hurricane. In the official 2021 file, a new arrival-without-dispatch state begins on 28 July. The raw audit shows a coordinated change across initiation streams: dispatch fields largely disappear from officer-initiated records while the same missing-dispatch configuration begins appearing among non-officer-initiated records. Ida occurs five weeks into this changed record regime and adds a temporary, unusually large reconfiguration.

The paper contributes four results. First, it binds the late-July break to the official monthly source bytes and shows that it is not an artifact of the aggregate pipeline. Second, it separates a common-support standardized comparison from a genuinely support-independent full-count comparison. Third, it uses public institutional documents to define what is and is not learned about the change: labels are clearer and two technology stories are inconsistent with the chronology, but the cause remains unidentified. Fourth, it translates the findings into a measurement rule for response-time and reported-domestic-violence research: declare the call denominator, public clock, endpoint coverage, and lineage before interpreting field durations as service performance.

Main finding. The released CAD measurement product changes before Ida and changes again during Ida. The public evidence identifies a record-production discontinuity and an event-window field reconfiguration. It does not identify physical dispatch, physical arrival, effective capacity, response quality, domestic-violence incidence, or the mechanism generating the public fields.

2 Related literature

2.1 Response time and first-responder performance

Response delays matter for crime clearance, victim harm, and public-service costs. Faster police arrival can improve clearance and reduce injuries, while congestion and staffing influence first-responder travel and queueing (Blanes i Vidal and Kirchmaier, 2018; DeAngelo et al., 2023; Brent and Beland, 2020; Mourtgos et al., 2024). A broader policing literature stresses that performance is multidimensional and that a single administrative clock may reward a selected stage rather than the public objective (Moore and Braga, 2004; Tiwana et al., 2015; Verlaan and Ruiter, 2023). This paper contributes a prerequisite: before estimating response-time determinants or consequences, establish that the public start and end fields describe a stable measurement object on a declared denominator.

2.2 Administrative records and measurement error

Administrative data are generated by institutions rather than designed solely for research. Recorded outcomes can reflect classification practice, verification, discretion, and incentives (Klinger and Bridges, 1997; Boivin and Cordeau, 2011; Boivin and Ouellet, 2014; Simpson and Orosco, 2021). Misclassification and measurement error can distort both group comparisons and structural interpretation (Kapteyn and Ypma, 2007; Chen et al., 2011; Knox et al., 2020). Government performance

systems can also alter the recorded object that is optimized or reported (Courty and Marschke, 1997; Eckhouse, 2022). The distinctive evidence here is topological: mutually exclusive patterns of public-field presence change sharply even before durations are analyzed.

2.3 Disasters and domestic violence

Disaster exposure, displacement, psychological stress, partner conflict, and service demand can move together (Anastario et al., 2009; Schumacher et al., 2010; Harville et al., 2011; First et al., 2022). Pandemic-era studies similarly show that calls, reports, and underlying victimization need not move one-for-one (Miller and Segal, 2019; Bullinger et al., 2021; Miller et al., 2024). These studies motivate attention to reported domestic-violence calls during emergencies, but they do not validate a particular CAD clock. Ida is used here as a system-level stress test of administrative observability, not as a treatment whose effect on domestic-violence incidence is estimated.

2.4 Partial identification and honest uncertainty

When the mapping from latent events to observed records is incomplete, point identification should not be assumed (Manski, 1989, 1990, 2003; Tamer, 2010; Molinari, 2020). Misclassified outcomes can instead support bounds under transparent restrictions (Horowitz and Manski, 1995; Molinari, 2008). That logic informs two choices here. Public-field configurations are analyzed directly without assigning them latent operational meanings, and downstream reported-DV measures retain $1/0/U$ classification and sharp bounds when classification or endpoint authority is unresolved.

3 Data and measurement

3.1 Official files and the declared denominator

The analysis uses the official annual New Orleans Calls for Service datasets for 2020–2024 and the official 2025 and 2026 files (City of New Orleans, 2021, 2025, 2026). The source audit binds each annual dataset identifier to twelve cached monthly files, observed date limits, row counts, and a deterministic SHA-256 bundle digest. The primary denominator is *non-officer-initiated public CAD records*, identified by the published `SelfInitiated` field. Officer-initiated records are a separate measurement stream, not extra observations silently pooled into the main denominator.

For record i , define $D_i = 1$ when the eventual released record has a nonblank public dispatch field and $A_i = 1$ when its arrival field parses and is no earlier than call creation. The four states are

$$J_{da,i} = \mathbf{1}\{D_i = d, A_i = a\}, \quad (d, a) \in \{0, 1\}^2. \quad (3)$$

Thus J_{01} means only that the released record contains a valid arrival field and no dispatch field. Calls are binned by creation time; stage fields are read from the eventual record. A later value may have been back-filled or overwritten.

The official metadata calls `TimeDispatch` the time entered when an officer was dispatched, `TimeArrive` the time entered when an officer arrived, and `SelfInitiated` an item generated by an officer in the field rather than one responding to 911 (City of New Orleans, 2021). These are administrative definitions. They do not disclose entry history, overwrite behavior, internal event identifiers, or the transformation from operational systems to the public export.

Table 1. Raw official-file audit around 28 July 2021

Date	Stream	Records	Dispatch present	Arrival valid	J_{01}
25 Jul.	Non-officer	721	631	532	0
26 Jul.	Non-officer	733	663	565	0
27 Jul.	Non-officer	731	658	585	0
28 Jul.	Non-officer	620	509	502	34
29 Jul.	Non-officer	666	510	532	69
30 Jul.	Non-officer	705	542	559	65
31 Jul.	Non-officer	725	584	572	45
27 Jul.	Officer	479	479	479	0
28 Jul.	Officer	575	230	575	345
29 Jul.	Officer	504	5	504	499
30 Jul.	Officer	548	3	548	545
31 Jul.	Officer	389	1	389	388

Dispatch present means nonblank `TimeDispatch`. Arrival valid means parseable `TimeArrive` no earlier than parseable `TimeCreate`. Every nonblank dispatch and arrival field in these rows is parseable. The table is aggregate and contains no record identifiers.

3.2 First-hand audit of the late-July break

The audit reads the official July 2021 compressed CSV directly. It records field nonblank status, timestamp parseability, valid arrival ordering, state counts, and format classes; no row-level values are retained. All nonblank dispatch and arrival fields in 25–31 July parse under the declared ISO rule, and no sentinel or format change explains the discontinuity.

The audit changes the interpretation in a useful but limited way. It verifies a real public-file discontinuity and shows that initiation streams change together. It makes an unnoticed aggregate-processing error implausible. It does not reveal whether the source was an internal workflow, field-entry convention, downstream transform, or export rule.

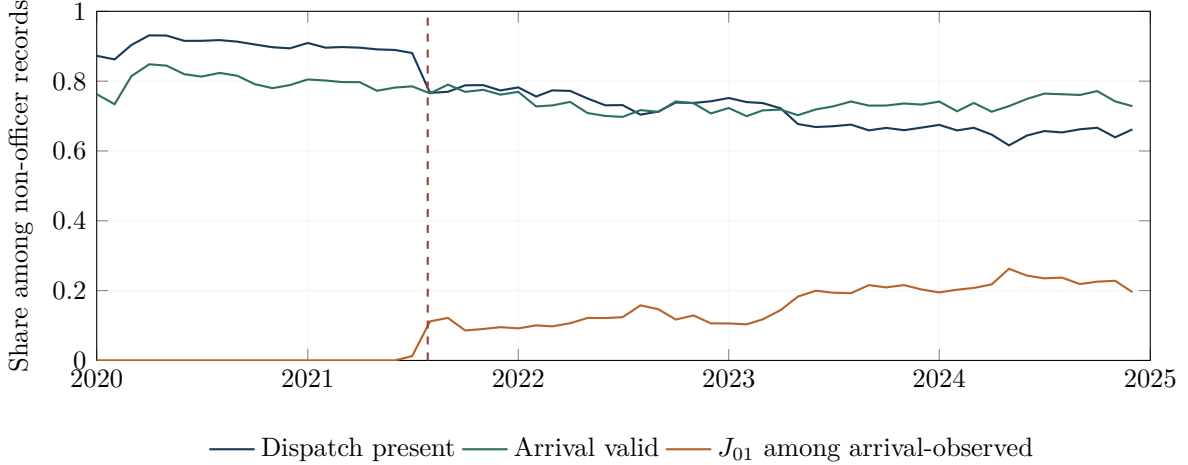


Figure 1. Monthly public-field completeness among non-officer-initiated records. The dashed line marks the first day with nonzero J_{01} . The figure describes released fields, not physical events.

4 Comparison design

The Ida event window is 29 August 00:00 through 3 September 00:00, divided into ten twelve-hour call-creation bins. Each bin is compared with the same half-day seven days earlier. The standardized matrix G contains event-minus-baseline contrasts in J_{01} , J_{10} , and J_{11} over common `initialtype` $\times$ `hour-bucket` $\times$ `schema-era` strata, weighted by baseline counts. The headline statistic is

$$M_{\max}(G) = \max_{t,j} |G_{tj}|. \quad (4)$$

The event and its matrix were known before the historical 0.25 sensitivity rule and restricted support library were completed. The three reference universes, eligibility rules, and context exclusions were fixed before the reference-window outcomes were inspected. The exercise is therefore a transparent descriptive calibration, not a preregistered randomization test.

The principal standardized universe contains 153 ordinary five-day windows starting on or after 1 July 2021. The standardized rule requires at least 0.90 common-support coverage on both event and baseline sides for reference windows. Ida is retained as the target even though it fails that rule, and its coverage is reported rather than concealed.

Two full-count comparisons use all non-officer records in each half-day. The first inherits the stage-era membership list; it removes common-support selection but remains tied to that universe. The second is genuinely support-independent with respect to the July break: both event and baseline sides must begin after 28 July, and prespecified emergency and holiday contexts remain excluded. This yields 151 post-change references.

5 Results

5.1 Ida is unusually large under both estimators

The standardized Ida statistic is $M_{\max} = 0.5072$, above all 153 stage-era references (rank 1/154 including Ida). The largest reference value is 0.3282. With 4,000 within-stratum multinomial replicates, Ida’s conditional 95-percent interval is $[0.460, 0.566]$; the strongest five reference-window intervals are reported in the supplement.

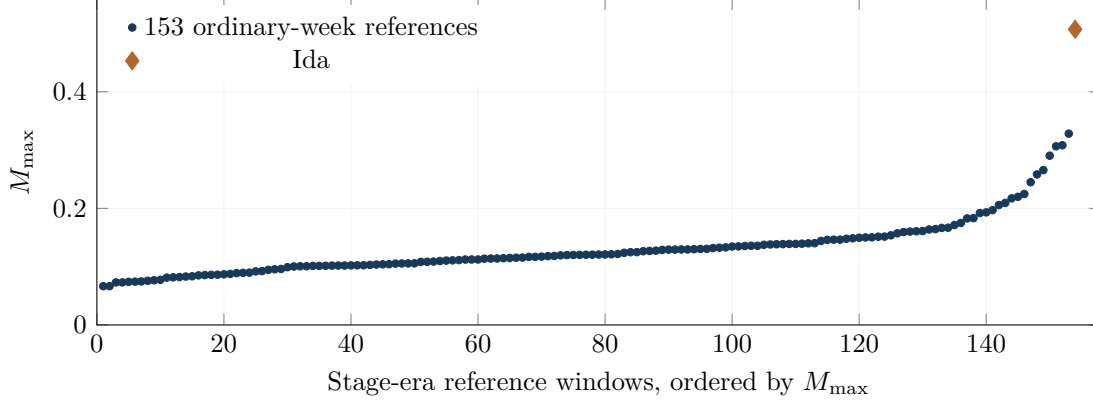


Figure 2. Maximum absolute standardized public-field contrast. The descriptive rank is not a causal or randomized p -value.

Table 2. Prespecified stage-era statistics

Statistic	Ida	Largest reference	Rank including Ida
$M_{\max}$ / amplitude A	0.5072	0.3282	1/154
σ_1	1.0837	0.6011	1/154
σ_2	0.2623	0.2451	1/154
σ_3	0.0858	0.1112	11/154
U_{direct}	0.4223	0.3282	1/154
U_{full} upper endpoint	0.2507	0.2125	1/154
Q upper endpoint	0.4944	0.9408	100/154
D upper endpoint	0.1716	0.2835	4/154

The all-record, full-count Ida statistic is 0.5320. It is rank 1 among 150 comparable members of the stage-era set and rank 1 among 151 genuinely post-change references (1/152 including Ida). Its bootstrap interval is $[0.475, 0.601]$. Recomputing Ida and all 151 references in 4,000 bootstrap draws places Ida first in every draw; the 2.5th, median, and 97.5th percentiles of its rank are all 1. This does not establish temporal exchangeability, but it shows that the rank is not driven by common-support selection or ordinary record-level sampling variation.

5.2 Standardized support is incomplete

Common-support coverage ranges from 0.748 to 0.928 on the event side and 0.505 to 0.851 on the baseline side. Record-weighted coverage is 0.861 for 3,878 event records and 0.770 for 3,332 baseline records. Ida therefore would not pass the 0.90 rule applied to candidate references. The standardized estimate describes a selected common-support population; the full-count comparison is the primary safeguard against that asymmetry.

The binary exceedance count is also threshold-sensitive. The full-set 95th-percentile maximum-cell threshold is 0.2094 and flags nine cells; the stage-era threshold is 0.2247 and flags eight; the same-season threshold is 0.2057 and flags nine. Continuous contrasts, coverage, and intervals are more informative than a single abnormal/normal label.

5.3 Not every statistic is extreme

Ida is first on the simple amplitude, the leading singular values, the direct restricted score, and the constrained LP score, but only 11th on the third singular value, 100th on Q , and fourth on D . The evidence is specifically a large public-field reconfiguration, not uniform extremeness under every summary. The LP remains robustness only: its lower endpoint clears the historical 0.25 rule by 0.000734, and 50 of 217 reference intervals have width above 0.01. A timing exercise is unfavorable because zero-padded shifts fit better than the documented institutional alignment. Neither calculation identifies a mechanism.

5.4 Disposition accounting and initiation streams

Among arrival-observed records, exact Kitagawa decompositions assign the late aggregate dispatch-field contrasts of -0.4950 and -0.4824 primarily to within-recorded-disposition changes (-0.5069 and -0.4946), offset by composition terms of $+0.0119$ and $+0.0123$. Changing observed final-disposition mix is therefore not the sole explanation. Selection into arrival observation, unobserved composition, semantic drift, and recording pathways remain possible.

The initiation-stream pattern is stronger after the raw audit. Officer-initiated dispatch completeness is 99.99 percent in 2020, 61.7 percent over 2021 because the break occurs in late July, and approximately 1 percent in each of 2022–2024. Arrival completeness remains about 99.8–99.9 percent. By contrast, non-officer dispatch completeness declines from 90.3 percent in 2020 to 65.4 percent in 2024. This joint change is consistent with a new public-recording convention linked to initiation status, but the public file does not reveal whether records were reclassified, different events were retained, or one timestamp was omitted in export.

6 What public institutional documents add

The institutional search changes the paper in three ways. First, the DataNOLA page and attached dictionary bind the released labels and provider. The same page warns that records are preliminary, may contain mechanical or human error, may change after investigation, and should not be used for comparison over time; its public changelog reports no archived changes (City of New Orleans, 2021). That warning does not explain the break, but it directly supports the measurement caution.

Second, the public OIPM Hurricane Ida report and emergency materials document the oversight context, outage concerns, curfew, and monitored NOPD activities (Office of the Independent Police Monitor, 2021). Public NOPD policy chapters describe intended communications, call response, and records practices. Yet the exact editions in force during July–September 2021 were not located for all relevant chapters, and policy statements do not reveal the historical machine-event to public-export transform. These documents are semantic and institutional context, not a bridge receipt.

Third, two candidate technology explanations are inconsistent with the public chronology. OPCD’s publicly indexed Carbyne APEX cutover is dated 24 June 2022, eleven months after the July 2021 discontinuity (Orleans Parish Communication District, 2022). The New Orleans Inspector General describes Hexagon OnCall Records as a records-management project that was contracted in 2020 but never launched, not as the public CAD export transition (New Orleans Office of Inspector General, 2024). Eliminating these candidates is useful negative evidence; it does not identify the remaining cause.

Institutional-evidence boundary

The public documents fill missing semantic, provenance, and exclusion evidence. They do not supply a versioned 2021 changelog, internal dispatch-event history, user-entry audit trail, retention rule, or public-export lineage. Mechanism identification therefore remains open.

7 Current data and denominator discipline

The longer series must compare like with like. In 2024, the non-officer dispatch-field share is 65.38 percent. In the full 2025 file, 136,712 of 207,050 non-officer records have dispatch fields (66.03 percent), compared with 1,117 of 122,720 officer-initiated records (0.91 percent). Pooling both streams gives 137,829 of 329,770 records, or 41.80 percent. The 2026 snapshot is similar: 66.84 percent among 130,346 non-officer records, 0.93 percent among 79,483 officer records, and 41.88 percent overall.

The comparable sequence is therefore 65.4 percent in 2024, 66.0 percent in 2025, and 66.8 percent in the 2026 snapshot. The overall 41.8 percent series answers a different question because it combines streams with sharply different recording conventions.

Table 3. Current dispatch-field completeness by denominator

Year	Denominator	Records	Dispatch present	Share
2024	Non-officer	212,314	138,804	65.38%
2025	Non-officer	207,050	136,712	66.03%
2025	Officer	122,720	1,117	0.91%
2025	All public rows	329,770	137,829	41.80%
2026 snapshot	Non-officer	130,346	87,129	66.84%
2026 snapshot	Officer	79,483	738	0.93%
2026 snapshot	All public rows	209,829	87,867	41.88%

8 Implications for reported domestic-violence research

Public CAD counts are reports created in an administrative system. They are not a direct measure of underlying domestic-violence incidence, victim need, or successful access to help. A defensible analysis should therefore separate four objects:

1. the declared reported-call denominator;
2. the classification rule, retaining $1/0/U$ when label authority is incomplete;
3. the public start and endpoint definitions with coverage by group and period; and
4. the lineage evidence connecting public fields to the target performance measure.

If n_1 records are definitely in the target class, n_0 definitely outside it, and n_U unresolved, then the class share lies in

$$\left[\frac{n_1}{n_1 + n_0 + n_U}, \frac{n_1 + n_U}{n_1 + n_0 + n_U} \right]. \quad (5)$$

Endpoint missingness similarly calls for full-denominator bounds or an explicit selected-stage estimand. Ida can motivate a stress test of these measurement objects, but it cannot by itself establish that domestic-violence incidence or police performance rose or fell.

9 Limitations and conclusion

The analysis uses eventual public records rather than internal event histories. The July break is date-localized in raw official bytes, but its institutional mechanism is not observed. Reference windows are descriptive and not exchangeable by design. The bootstrap is conditional on observed strata and does not capture temporal dependence or uncertainty in the regime boundary. Holidays, day-of-week patterns, heat, weather, COVID-era behavior and mobility, reporting and help-seeking thresholds, redaction and suppression, and changing call composition can affect comparisons. Public `SelfInitiated=N` is not equivalent to “911 only.”

The central conclusion is constructive. Public CAD is valuable when its observable object is stated precisely. In New Orleans, field-presence topology identifies a measurement-regime change before Ida and a further Ida-era stress. Raw audit, denominator repair, source lineage, bootstrap expansion, and institutional documents make that conclusion more secure. They also sharpen what remains missing: the historical bridge from internal events and entry histories to the released fields. Until that bridge is observed, public timestamps should be reported as administrative-field measures rather than treated automatically as operational ground truth.

References

- Anastario, M., Shehab, N., and Lawry, L. (2009). Increased gender-based violence among women internally displaced in mississippi 2 years post-hurricane katrina. *Disaster Medicine and Public Health Preparedness*, 3(1):18–26.
- Blanes i Vidal, J. and Kirchmaier, T. (2018). The effect of police response time on crime clearance rates. *The Review of Economic Studies*, 85(2):855–891.
- Boivin, R. and Cordeau, G. (2011). Measuring the impact of police discretion on official crime statistics: A research note. *Police Quarterly*, 14(2):186–203.
- Boivin, R. and Ouellet, F. (2014). Space and time variations in crime-recording practices within a large municipal police agency. *International Journal of Police Science & Management*, 16(3):171–183.
- Brent, D. and Beland, L.-P. (2020). Traffic congestion, transportation policies, and the performance of first responders. *Journal of Environmental Economics and Management*, 103:102339.
- Bullinger, L. R., Carr, J. B., and Packham, A. (2021). COVID-19 and crime: Effects of stay-at-home orders on domestic violence. *American Journal of Health Economics*, 7(3):249–280.
- Chen, X., Hong, H., and Nekipelov, D. (2011). Nonlinear models of measurement errors. *Journal of Economic Literature*, 49(4):901–937.
- City of New Orleans (2021). Calls for service 2021. Data.NOLA dataset 3pha-hum9; official page, disclaimer, attached data dictionary, and empty archived changelog verified 25 August 2026.
- City of New Orleans (2025). Calls for service 2025. Data.NOLA dataset 4xwx-sfte; official page verified 24 August 2026.
- City of New Orleans (2026). Calls for service 2026. Data.NOLA dataset es9j-6y5d; official page verified 24 August 2026.
- Courty, P. and Marschke, G. (1997). Measuring government performance: Lessons from a federal job-training program. *American Economic Review*, 87(2):383–388.
- DeAngelo, G., Toger, M., and Weisburd, S. (2023). Police response time and injury outcomes. *The Economic Journal*, 133(654):2147–2177.
- Eckhouse, L. (2022). Metrics management and bureaucratic accountability: Evidence from policing. *American Journal of Political Science*, 66(2):385–401.
- First, J. M., Ravi, K. E., Smith-Frigerio, S., and Houston, J. B. (2022). Mental health impacts of hurricane harvey: Examining the roles of intimate partner violence and resilience. *Social Work Research*, 46(4):293–303.
- Harville, E. W., Taylor, C. A., Tesfai, H., Xiong, X., and Buekens, P. (2011). Experience of hurricane katrina and reported intimate partner violence. *Journal of Interpersonal Violence*, 26(4):833–845.

- Horowitz, J. L. and Manski, C. F. (1995). Identification and robustness with contaminated and corrupted data. *Econometrica*, 63(2):281–302.
- Kapteyn, A. and Ypma, J. Y. (2007). Measurement error and misclassification: A comparison of survey and administrative data. *Journal of Labor Economics*, 25(3):513–551.
- Klinger, D. A. and Bridges, G. S. (1997). Measurement error in calls-for-service as an indicator of crime. *Criminology*, 35(4):705–726.
- Knox, D., Lowe, W., and Mummolo, J. (2020). Administrative records mask racially biased policing. *American Political Science Review*, 114(3):619–637.
- Manski, C. F. (1989). Anatomy of the selection problem. *Journal of Human Resources*, 24(3):343–360.
- Manski, C. F. (1990). Nonparametric bounds on treatment effects. *American Economic Review*, 80(2):319–323.
- Manski, C. F. (2003). *Partial Identification of Probability Distributions*. Springer, New York.
- Miller, A. R. and Segal, C. (2019). Do female officers improve law enforcement quality? effects on crime reporting and domestic violence. *The Review of Economic Studies*, 86(5):2220–2247.
- Miller, A. R., Segal, C., and Spencer, M. K. (2024). Effects of the COVID-19 pandemic on domestic violence in los angeles. *Economica*, 91(361):163–187.
- Molinari, F. (2008). Partial identification of probability distributions with misclassified data. *Journal of Econometrics*, 144(1):81–117.
- Molinari, F. (2020). Microeconometrics with partial identification. In *Handbook of Econometrics*, volume 7A, pages 355–486. Elsevier.
- Moore, M. H. and Braga, A. A. (2004). Police performance measurement: A normative framework. *Criminal Justice Ethics*, 23(1):3–19.
- Mourtgos, S. M., Adams, I. T., and Nix, J. (2024). Staffing levels are the most important factor influencing police response times. *Policing: A Journal of Policy and Practice*, 18:paae002.
- New Orleans Office of Inspector General (2024). The orleans parish communication district hexagon project: In brief. Technical report, City of New Orleans. Official report concerning the unlaunched OnCall Records project.
- Office of the Independent Police Monitor (2021). 2021 hurricane season police oversight report: Understanding hurricane ida and policing response. Public institutional report, verified 25 August 2026.
- Orleans Parish Communication District (2022). Ribbon cutting to celebrate cut over to AT&T ESInet and Carbyne APEX call handling system. Official page indexed with a 24 June 2022 cutover date; the page returned not found when checked 25 August 2026.

- Schumacher, J. A., Coffey, S. F., Norris, F. H., Tracy, M., Clements, K., and Galea, S. (2010). Intimate partner violence and hurricane katrina: Predictors and associated mental health outcomes. *Violence and Victims*, 25(5):588–603.
- Simpson, R. and Orosco, C. (2021). Re-assessing measurement error in police calls for service: Classifications of events by dispatchers and officers. *PLOS ONE*, 16(12):e0260365.
- Tamer, E. (2010). Partial identification in econometrics. *Annual Review of Economics*, 2:167–195.
- Tiwana, N., Bass, G., and Farrell, G. (2015). Police performance measurement: An annotated bibliography. *Crime Science*, 4(1):1.
- Verlaan, T. and Ruiter, S. (2023). Predictors of police response time: A scoping review. *Crime Science*, 12:19.

Online Empirical and Methods Supplement

Public CAD Is Not Operational Ground Truth

25 August 2026

Scope

This supplement documents the public-source lineage, raw July audit, comparison design, support diagnostics, resampling, initiation-stream evidence, current-data denominator audit, institutional-source check, and reproduction contract. It is organized around the empirical claims in the main paper. Extended identified-set proofs, constrained-program certificates, and formal-verification material are kept in the separate *Legacy Mathematical and Formal-Verification Archive* and are not appended to the professor-facing paper.

Contents

1	Claim and evidence hierarchy	2
2	Annual public-source lineage	2
3	Raw July audit	3
4	State matrix and estimator	4
5	Common-support audit	4
6	Bootstrap expansion	5
7	Thresholds, secondary statistics, and excluded episodes	6
8	Officer-initiation evidence	7
9	Disposition accounting	8
10	Current-data denominator audit	9
11	Institutional-source audit	9
12	Reported-DV measurement implications	9
13	Reproduction and machine-consistency contract	10
14	Interpretive checklist	10
15	Numerical claim index	12

1 Claim and evidence hierarchy

The analysis distinguishes four evidence classes.

1. **First-hand public bytes:** official DataNOLA CSVs and their hashes.
2. **Deterministic aggregate transforms:** daily, half-day, monthly, and annual state counts derived from those files.
3. **Institutional documents:** official metadata, data dictionary, policy, oversight, and audit reports that define labels or exclude candidate explanations.
4. **Latent operational objects:** internal dispatch events, queue states, availability, entry histories, overwrite logs, and export transformations, which are not observed.

The first three classes identify a public-record regime change and delimit candidate interpretations. They do not turn the fourth class into observed data. Accordingly, J_{01} is always a public configuration—arrival field present, dispatch field absent—rather than a claim of physical arrival without dispatch.

2 Annual public-source lineage

2.1 Dataset binding

Table 1 binds each analysis year to the official DataNOLA identifier and the first-hand cache audit. Each annual bundle hash is computed by sorting the twelve monthly filenames and hashing the filename plus each file’s SHA-256 digest. This avoids concatenating or redistributing the raw source files. The monthly cache did not retain a reliable acquisition-date field, so the table reports that limitation rather than converting filesystem timestamps into retrieval dates. All sources were reverified on 25 August 2026.

Table 1. Annual DataNOLA source lineage, 2020–2024

Year	Dataset	Observed creation range	Rows	Files	Annual bundle SHA-256
2020	hp7u-i9hf	1 Jan.–31 Dec. 2020	432,892	12	a1d3d7fffb6a34e7...da7cc0db1
2021	3pha-hum9	1 Jan.–31 Dec. 2021	428,315	12	53b0ae39abbbc363...a7edc559
2022	nci8-thrr	1 Jan.–31 Dec. 2022	354,207	12	1ca797cd30109a80...e3d1e4d8
2023	pc5d-tvaw	1 Jan.–31 Dec. 2023	325,091	12	68d583fe0c4dbb43...4f011a13a
2024	2zcj-b6ts	1 Jan.–31 Dec. 2024	327,696	12	041f7ac52458c797...b4f6f464

Exact observed minimum and maximum timestamps, full hashes, monthly-file hashes, byte counts, local cache mtimes, official URLs, parser version, and verification date are in `r15_annual_source_lineage.csv` and `r15_public_source_audit.json`. Retrieval dates were not recorded in the monthly cache.

The annual row counts are raw public-file rows before the paper’s initiation restriction. Analysis denominators are constructed after parsing `SelfInitiated`. No narratives or addresses are used. Item identifiers are used only by the upstream canonical pipeline for deduplication and are not written to review artifacts.

2.2 Official metadata and changelog

The official 2021 page identifies OPCD as data provider, reports 21 public columns, supplies a data-dictionary attachment, and warns about preliminary classifications, later changes, mechanical or human error, sequencing, and over-time comparisons (City of New Orleans, 2021). When

inspected, its public “Dataset Changelog” stated that no changes had been archived. This is itself an observability limitation: the public page does not provide a dated change record that could explain 28 July 2021.

3 Raw July audit

3.1 Parsing and state construction

The audit applies the same public-field definitions as the aggregate analysis:

$$D_i = \mathbf{1}\{\text{TimeDispatch}_i \text{ is nonblank}\}, \quad (1)$$

$$A_i = \mathbf{1}\{\text{TimeArrive}_i \text{ parses and is no earlier than TimeCreate}_i\}. \quad (2)$$

Timestamps are parsed as local ISO strings with a T or single-space separator. Empty strings and the textual values `null`, `none`, `nan`, and `nat` are missing sentinels. All nonblank dispatch and arrival values in the audit interval parse. Format-class counts show no switch that could produce the break mechanically.

Table 2. Complete aggregate audit, 25–31 July 2021

Date	Stream	N	D nonblank	D parsed	A nonblank	A valid	J_{00}	J_{10}	J_{01}	J_{11}
25 Jul.	Non-officer	721	631	631	532	532	90	99	0	532
26 Jul.	Non-officer	733	663	663	565	565	70	98	0	565
27 Jul.	Non-officer	731	658	658	585	585	73	73	0	585
28 Jul.	Non-officer	620	509	509	502	502	77	41	34	468
29 Jul.	Non-officer	666	510	510	532	532	87	47	69	463
30 Jul.	Non-officer	705	542	542	559	559	98	48	65	494
31 Jul.	Non-officer	725	584	584	572	572	96	57	45	527
25 Jul.	Officer	377	377	377	377	377	0	0	0	377
26 Jul.	Officer	404	404	404	404	404	0	0	0	404
27 Jul.	Officer	479	479	479	479	479	0	0	0	479
28 Jul.	Officer	575	230	230	575	575	0	0	345	230
29 Jul.	Officer	504	5	5	504	504	0	0	499	5
30 Jul.	Officer	548	3	3	548	548	0	0	545	3
31 Jul.	Officer	389	1	1	389	389	0	0	388	1

3.2 What the audit resolves

The audit resolves three concerns. The first J_{01} date is present in official source bytes. The change survives a parseability audit. And the abrupt officer-stream shift shows that the non-officer pattern is part of a broader public-record transition rather than an isolated fluctuation in one classified group.

The audit does not resolve the institutional cause. It has no event-entry history, user action, intermediate CAD object, export log, or versioned retention rule. The strongest supported wording is therefore “public-recording transition associated with initiation status,” not a named software or

operational mechanism.

4 State matrix and estimator

For half-day bin t , state $j \in \{J_{01}, J_{10}, J_{11}\}$, and common stratum x , let p_{tjx}^E and p_{tjx}^B be event and baseline shares and let w_{tx}^B be the baseline weight over common support. The standardized contrast is

$$G_{tj} = \sum_{x \in \mathcal{X}_t^E \cap \mathcal{X}_t^B} w_{tx}^B (p_{tjx}^E - p_{tjx}^B). \quad (3)$$

The simple amplitude is $M_{\max} = \max_{t,j} |G_{tj}|$. The event window begins 29 August 2021 and contains ten twelve-hour bins; the baseline is seven days earlier.

The standardized stage-era universe consists of 153 retained ordinary-week references beginning on or after 1 July 2021. Sixty-six eligible full-universe references occur entirely before 28 July, when J_{01} is structurally absent, and 64 start before the 1 July stage cutoff. These facts explain why a common formula does not imply a common measurement regime.

The full-count statistic drops the stratum intersection and weights. For each half-day it compares unconditional state shares and again takes the maximum absolute cell contrast. Two reference sets are reported:

- the inherited stage-era membership set, which removes support selection but retains the existing membership rule; and
- a post-change set requiring both event and baseline sides to begin after 28 July, while retaining the frozen context exclusions.

The second is the support-independent comparison for the post-change measurement product.

5 Common-support audit

The minimum coverages are 0.748 and 0.505; record-weighted coverages are 0.861 and 0.770. The symmetric 0.90 rule fails. Consequently, the standardized result cannot be described as the full event-window population. The full-count statistic uses all 3,878 event and 3,332 baseline records.

Table 3. Ida common-support coverage and record counts

Bin	Event N	Baseline N	Event coverage	Baseline coverage
B1	276	244	0.928	0.754
B2	674	335	0.901	0.612
B3	206	291	0.748	0.505
B4	495	433	0.919	0.815
B5	295	290	0.766	0.838
B6	464	421	0.873	0.812
B7	311	268	0.846	0.802
B8	464	379	0.851	0.836
B9	240	255	0.829	0.812
B10	453	416	0.834	0.851

6 Bootstrap expansion

6.1 Standardized windows

Within each observed stratum, records are resampled as multinomial draws over $(J_{00}, J_{10}, J_{01}, J_{11})$ while holding the observed stratum totals and baseline weights fixed. This is exactly the record-level nonparametric bootstrap available from the aggregate state counts. It captures record sampling conditional on the observed strata; it does not capture serial dependence, regime-boundary uncertainty, or design selection.

6.2 Full-count windows and rank distribution

The same multinomial logic is applied to unconditional half-day state counts. Ida’s full-count interval is $[0.4754, 0.6015]$. The strongest post-change reference point is 0.3098 with interval $[0.2599, 0.3799]$; the remaining top-five intervals have upper endpoints between 0.3636 and 0.3843. For each of 4,000 draws, Ida is ranked against all 151 independently resampled post-change references. Rank 1 occurs in 100 percent of draws, and the reported rank interval is $[1, 1]$.

Table 4. Bootstrap intervals for Ida and the five strongest standardized references

Window	Point	2.5th percentile	97.5th percentile
Ida, 29 Aug. 2021	0.5072	0.4597	0.5656
15 May 2022	0.3282	0.2655	0.3954
22 May 2022	0.3083	0.2528	0.3725
12 May 2024	0.3064	0.2502	0.3880
5 May 2024	0.2904	0.2355	0.3660
10 Dec. 2023	0.2656	0.1630	0.3764

Table 5. Full-count bootstrap: Ida and five strongest post-change references

Window	Point	2.5th percentile	97.5th percentile
Ida, 29 Aug. 2021	0.5320	0.4754	0.6015
15 May 2022	0.3098	0.2599	0.3799
10 Dec. 2023	0.2978	0.2140	0.3843
19 Jun. 2022	0.2959	0.2384	0.3636
5 May 2024	0.2941	0.2501	0.3735
12 May 2024	0.2926	0.2238	0.3773

7 Thresholds, secondary statistics, and excluded episodes

The 95th-percentile maximum-cell thresholds are 0.2094 in the full set, 0.2247 in the stage-era set, and 0.2057 in the same-season set. They yield 9, 8, and 9 exceeding cells. Reporting one threshold without the other two would overstate the stability of a binary label.

The constrained LP is a robustness calculation. Its Ida lower endpoint exceeds 0.25 by 0.000734, while 50 of 217 reference intervals have width above 0.01 and the maximum width is 0.070. The numerical LP status is therefore a tested numerical result, not an exact rational certificate. Formal code verifies arithmetic and finite certificate logic; it does not verify the empirical meaning of public fields.

Excluded episodes are not members of the ordinary-week rank. Pre-break events cannot reproduce a positive J_{01} contrast because that state is absent from their public regime. The large post-Ida statistic compares a recovery week with Ida itself. Francine is much smaller in the later regime. These comparisons reinforce the need to distinguish event stress from the measurement object used to observe it.

Table 6. Secondary stage-era statistics

Statistic	Ida	Largest reference	Rank including Ida
$M_{\max} / A$	0.5072	0.3282	1/154
σ_1	1.0837	0.6011	1/154
σ_2	0.2623	0.2451	1/154
σ_3	0.0858	0.1112	11/154
U_{direct}	0.4223	0.3282	1/154
$U_{\text{full upper}}$	0.2507	0.2125	1/154
Q_{upper}	0.4944	0.9408	100/154
D_{upper}	0.1716	0.2835	4/154

Table 7. Labelled excluded episodes: full-count maximum-cell contrast

Episode window	Event records	Baseline records	Raw $M_{\max}$
Hurricane Laura, 23 Aug. 2020	3,316	3,515	0.084
Hurricane Zeta, 25 Oct. 2020	3,963	3,638	0.129
February freeze, 14 Feb. 2021	2,851	3,329	0.113
Hurricane Ida, 29 Aug. 2021	3,878	3,332	0.532
Post-Ida week, 5 Sep. 2021	3,827	3,878	0.478
Hurricane Francine, 8 Sep. 2024	2,886	2,882	0.110

8 Officer-initiation evidence

The main analysis excludes officer-initiated records because their public dispatch convention differs radically. Their pattern is nevertheless analytically valuable. It shows that J_{01} is a normal public configuration in one stream after the break and that pooling initiation streams creates a denominator artifact. It does not prove that non-officer calls were operationally converted into officer activity.

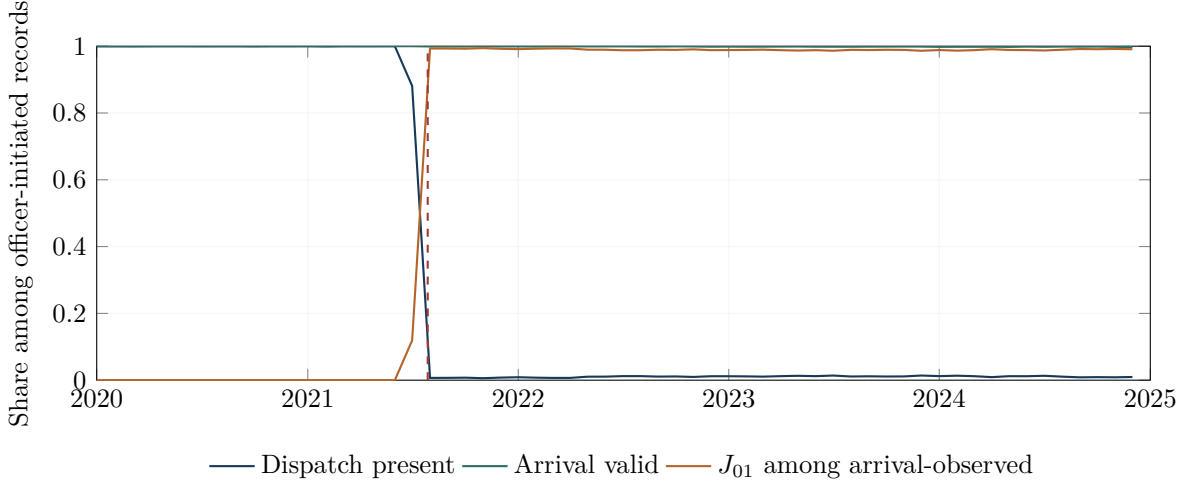


Figure 1. Officer-initiated public-field pattern, 2020–2024. Dispatch presence moves from essentially complete to approximately 1 percent while arrival remains essentially complete.

Table 8. Annual field completeness by initiation stream

Year	Stream	Records	Dispatch share	Arrival share
2020	Non-officer	271,868	0.903	0.801
2021	Non-officer	258,160	0.845	0.783
2022	Non-officer	243,179	0.745	0.724
2023	Non-officer	225,389	0.693	0.722
2024	Non-officer	212,314	0.654	0.742
2020	Officer	161,024	1.000	1.000
2021	Officer	170,155	0.617	1.000
2022	Officer	111,028	0.010	0.999
2023	Officer	99,702	0.012	0.999
2024	Officer	115,382	0.011	0.998

9 Disposition accounting

Among arrival-observed records, let p_{dt} be the recorded-disposition share and q_{dt} the dispatch-field rate within disposition d at time t . The aggregate change decomposes exactly as

$$\Delta \bar{q} = \sum_d \bar{p}_d \Delta q_d + \sum_d \bar{q}_d \Delta p_d, \quad (4)$$

where bars denote midpoints. For the two largest late Ida contrasts, the totals -0.4950 and -0.4824 decompose into within-disposition terms -0.5069 and -0.4946 plus composition terms $+0.0119$ and $+0.0123$. The observed final-disposition mix is not sufficient to explain the shift. Because the sample conditions on arrival observation and uses final recorded dispositions, the decomposition does not identify operational routing or causal mechanisms.

Table 9. 2025 and 2026 snapshot field completeness

Year	Denominator	Records	Dispatch present	Dispatch share
2025	Non-officer	207,050	136,712	0.6603
2025	Officer	122,720	1,117	0.0091
2025	All rows	329,770	137,829	0.4180
2026 snapshot	Non-officer	130,346	87,129	0.6684
2026 snapshot	Officer	79,483	738	0.0093
2026 snapshot	All rows	209,829	87,867	0.4188

10 Current-data denominator audit

The 2025 all-row rate is arithmetically correct but scientifically noncomparable to the non-officer series. The comparable series is 0.6538 (2024), 0.6603 (2025), and 0.6684 (2026 snapshot). The overall rate is low because officer-initiated records have nearly universal arrival fields and almost no dispatch fields under the later public convention.

11 Institutional-source audit

Table 10. Public institutional documents and what they identify

Source	Public status	What it adds	What it does not add
DataNOLA 2021 page and dictionary	Official page and attachment public	Field labels, provider, schema, preliminary-data warning, no archived changelog	Entry history, overwrite rule, internal/public lineage, July change cause
OIPM Hurricane Ida report	Official PDF public	Oversight context, emergency conditions, monitored activities	Machine-event semantics, export history, causal explanation of J states
NOPD manual chapters	Public editions located	Intended communications, call-response, and records practices	Complete exact-edition set for July–September 2021; proof of implementation
OPCD Carbyne APEX notice	Official page indexed; current URL returns not found	Cutover date of 24 June 2022	Explanation of July 2021 break
OIG Hexagon report	Official PDF public	OnCall Records project history; system was not launched	Evidence of a CAD-export transition

These documents supply genuine missing pieces: source semantics, public warnings, chronology, and negative evidence against two named technology stories. They still leave the decisive bridge missing. A mechanism claim would require a versioned changelog or internal/public reconciliation showing how a particular event and entry history became the released row.

12 Reported-DV measurement implications

The paper conditions on reported calls created in CAD. It does not observe latent need, attempts to seek help, contacts not converted into CAD items, or underlying domestic-violence incidence.

Broad text or code rules can also create false positives, and public suppression can create false negatives. The recommended classification remains $1/0/U$ until an authoritative crosswalk resolves the uncertain group.

If n_1 , n_0 , and n_U are the definitely included, definitely excluded, and unresolved counts, then the sharp no-assumption prevalence bounds are

$$\frac{n_1}{n_1 + n_0 + n_U} \leq p \leq \frac{n_1 + n_U}{n_1 + n_0 + n_U}. \quad (5)$$

For response-time outcomes, endpoint coverage must be stated on the full declared denominator. A duration among rows with both timestamps is a selected-stage statistic unless missing endpoints are bounded or independently qualified. The system-level Ida stress test shows why these requirements are substantive rather than cosmetic.

13 Reproduction and machine-consistency contract

The package build performs the following sequence:

1. rebuild aggregate diagnostics and all tables from the packaged frozen tally and reference artifacts;
2. optionally rerun the raw-source lineage audit against the canonical official cache;
3. compile the main paper, this supplement, research status note, and separate legacy archive;
4. combine only the main paper and empirical supplement into the professor-facing PDF;
5. check exact headline values, rank denominators, support status, threshold counts, 2025/2026 denominator arithmetic, source-audit invariants, and forbidden revision-history language in the main source;
6. verify PDF names, page relationships, text presence, manifest hashes, and the frozen reproduction snapshot.

The ordinary builder uses Python’s standard library and NumPy. SciPy is required only by the inherited constrained-optimization reproduction path, and pypdf is used to combine PDFs. Pandas is not required. The package manifest covers all curated files except the manifest itself and any generated zip. Text normalization performed by packaging is checked through the rebuilt numerical and PDF parity tests rather than treated as byte identity with an external working tree.

14 Interpretive checklist

Before using a public CAD duration as a performance measure, report:

1. the source identifier, observed date range, row count, and source hash;
2. the declared denominator and initiation-stream rule;
3. the exact start and endpoint semantics;
4. field nonblank, parseability, and ordering rates;
5. missing-endpoint handling and support coverage;
6. versioned institutional lineage, if available;
7. whether the target is a released-field statistic, selected-stage measure, bounded performance functional, or unavailable latent object;
8. shock exclusions, day-of-week, holidays, heat, weather, seasonality, COVID-era behavior, re-

porting behavior, help-seeking, suppression, and classification uncertainty.

References

City of New Orleans (2021). Calls for service 2021. Data.NOLA dataset 3pha-hum9; official page, disclaimer, attached data dictionary, and empty archived changelog verified 25 August 2026.

15 Numerical claim index

Table 11 identifies the package object that supplies each headline number. The release verifier recomputes the claims from these machine-readable objects; the table is an orientation aid, not an independent evidentiary source.

Table 11. Headline claims and their machine-readable sources

Claim	Reported value	Primary package object
Standardized Ida maximum	0.5072; rank 1/154	<code>r15_aggregate_diagnostics.json</code>
Full-count Ida maximum	0.5320; rank 1/151	<code>r15_raw_window_scores.csv</code>
Support-independent post-change rank	1/152 including Ida	<code>r15_aggregate_diagnostics.json</code>
Threshold exceedances	9, 8, and 9	<code>r15_aggregate_diagnostics.json</code>
Standardized bootstrap interval	[0.4597, 0.5656]	<code>r15_bootstrap_window_maxima.csv</code>
Full-count bootstrap interval	[0.4754, 0.6015]	<code>r15_bootstrap_window_maxima.csv</code>
2025 non-officer dispatch share	$136,712/207,050 = 0.6603$	<code>r15_current_denominator_audit.csv</code>
2025 officer dispatch share	$1,117/122,720 = 0.0091$	<code>r15_current_denominator_audit.csv</code>
First non-officer J_{01} day	28 July 2021	<code>r15_raw_july_25_31_audit.csv</code>
Annual public-source bindings	five official dataset IDs	<code>r15_annual_source_lineage.csv</code>

The aggregate objects are fully contained in the reproduction snapshot. The raw July and annual-lineage objects add first-hand source checks against the canonical public cache. Institutional documents remain a separate evidence class: they inform field semantics, warnings, and chronology, but do not generate or alter the numerical statistics above.